

Homotopy-Preserving Logical Observations in Propositional Logical Topologies

Repair maps, closure systems, Horn-type criteria, and the height-two repair-forest theorem

Working research note v0.4 — 23 September 2026

Continuation of *Homological Foundations for Propositional Logical Topologies* (v0.1), *Observation Refinement Classification* (v0.2), and *Controlled Refinement Regimes* (v0.3).

Status convention. [S] denotes standard mathematics used essentially unchanged; [A] denotes a direct ProLT specialization or elementary derived result; [C] denotes a contribution candidate whose novelty is not claimed before a dedicated adversarial literature audit. The strongest new candidate in this note is the exact height-two *repair-forest theorem*, which converts one-bit post- T_0 refinement into a rooted graph and makes homotopy preservation equivalent to a tree/forest condition.

1 Purpose and relation to the previous stages

The ProLT core manuscript starts with an indexed observation family Φ , its signature map o_Φ , and the finite topology generated by positive truth regions. Its key order-theoretic fact is

$$p \preceq_\Phi q \iff o_\Phi(p) \subseteq o_\Phi(q).$$

The v0.1 note then defined canonical ProLT homology using the order complex of the realized-signature poset. The v0.2 note classified arbitrary refinements through their new-coordinate fiber profiles and introduced mapping-cone defect homology. The v0.3 note specialized to post- T_0 refinement, where no state splitting remains: refinement keeps the same vertices and only deletes old order comparisons. It proved, among other results:

- (i) positive/isotone added observations are exactly the topologically redundant one-bit observations;
- (ii) nonconstant antitone added observations split connected components and change H_0 ;
- (iii) height at most one is rigid: every nonredundant post- T_0 refinement changes homology;
- (iv) height two admits an exact relative-boundary-matrix test;
- (v) over a coarse chain, one added bit is homotopy-defective exactly for a nonconstant nonincreasing threshold word.

The present goal is more operational:

Which logically recognizable kinds of new observations are guaranteed to preserve the homotopy type of the canonical ProLT order complex?

Two answers are developed. First, in arbitrary height, we obtain constructive *repair-map* criteria, with closure-system and Horn-type corollaries. Second, in height two, the previous relative matrix becomes a graph incidence matrix, producing an exact rooted-forest characterization and an explicit greedy collapse algorithm.

2 Post- T_0 one-bit refinement

Assume throughout this section that the coarse ProLT is already T_0 . Write

$$P := P_\Phi$$

for its realized-signature poset. Let ψ be one new observation and define

$$b(p) := 1[p \models \psi] \in \{0, 1\}.$$

The refined poset has the same underlying set and order

$$p \leq_b q \iff p \leq q \text{ in } P \text{ and } b(p) \leq b(q). \quad (1)$$

Write

$$Q_b := (P, \leq_b), \quad K := \Delta(P), \quad L_b := \Delta(Q_b) \subseteq K.$$

Definition 2.1 (Descent). A comparable pair $p < q$ is a b -descent when

$$b(p) = 1, \quad b(q) = 0.$$

Equivalently, $[p, q]$ is an edge of K deleted by the refinement.

Proposition 2.2 (Deleted-chain criterion). *[A] A coarse chain*

$$p_0 < p_1 < \dots < p_k$$

survives as a simplex of L_b if and only if

$$b(p_0) \leq b(p_1) \leq \dots \leq b(p_k).$$

Thus a simplex is deleted exactly when its Boolean word contains a $1 \rightarrow 0$ descent.

This makes the logical content of refinement very concrete: after T_0 , a new Boolean observation deletes precisely those information chains along which the new truth value reverses from true to false.

3 Repair maps: arbitrary-height sufficient criteria

The simplest way to prove that $L_b \hookrightarrow K$ is a homotopy equivalence is to construct a monotone map that systematically moves every old state to a “safe” state without leaving its old information direction.

3.1 General repair-map theorem

Theorem 3.1 (Down-repair criterion). *[S]/[A]/[C] Suppose there exists an order-preserving map*

$$\rho : P \longrightarrow P$$

such that, for every $p \in P$,

$$\rho(p) \leq p, \quad b(\rho(p)) = 0. \quad (2)$$

Then the inclusion

$$i : Q_b \hookrightarrow P$$

induces a homotopy equivalence

$$\Delta(Q_b) \simeq \Delta(P).$$

The map ρ , regarded as a map $P \rightarrow Q_b$, is a homotopy inverse.

Proof. Because ρ is monotone in P and every value of $b \circ \rho$ is 0, the map $\rho : P \rightarrow Q_b$ is monotone. Moreover,

$$i\rho(p) = \rho(p) \leq p$$

in P . For $p \in Q_b$, condition (2) also gives

$$\rho(p) \leq_b p,$$

because $0 = b(\rho(p)) \leq b(p)$. Hence

$$i\rho \leq \text{id}_P, \quad \rho i \leq \text{id}_{Q_b}.$$

Pointwise comparable monotone maps induce homotopic maps on order complexes, so $\Delta(i)$ and $\Delta(\rho)$ are homotopy inverses. $\square$

Theorem 3.2 (Up-repair criterion). *[S]/[A]/[C] Dually, suppose there exists an order-preserving map*

$$\lambda : P \longrightarrow P$$

such that

$$p \leq \lambda(p), \quad b(\lambda(p)) = 1$$

for every p . Then $L_b \hookrightarrow K$ is a homotopy equivalence, with $\lambda : P \rightarrow Q_b$ as a homotopy inverse.

Remark 3.3 (Interpretation). A down-repair monotonically weakens every information state to a ψ -false state. An up-repair monotonically strengthens every state to a ψ -true state. Either construction routes every state through a region on which the new observation has constant truth value, so the newly forbidden true-to-false comparisons can be bypassed without changing homotopy type.

3.2 Canonical false floors and true ceilings

Let

$$F_\psi := \{p : b(p) = 0\}, \quad T_\psi := \{p : b(p) = 1\}.$$

Definition 3.4 (False-floor property). The observation ψ has the *false-floor property* when, for every $p \in P$, the set

$$F_\psi \cap \downarrow p$$

has a greatest element, denoted $\lfloor p \rfloor_F$.

Definition 3.5 (True-ceiling property). The observation ψ has the *true-ceiling property* when, for every $p \in P$, the set

$$T_\psi \cap \uparrow p$$

has a least element, denoted $\lceil p \rceil_T$.

Theorem 3.6 (Floor/ceiling preservation). *[S]/[A]/[C] If ψ has either the false-floor property or the true-ceiling property, then adding ψ preserves the homotopy type of the canonical ProLT order complex.*

Proof. For the false-floor case define

$$\rho(p) = \lfloor p \rfloor_F.$$

If $p \leq q$, then every false state below p is also below q , so maximality gives

$$\rho(p) \leq \rho(q).$$

Thus ρ is monotone, lies below the identity, and takes only false values. Apply Theorem 3.1. The true-ceiling case is dual. $\square$

Remark 3.7 (Closure-operator form). The floor map is an interior operator: it is monotone, idempotent, and satisfies $\rho(p) \leq p$. The ceiling map is a closure operator: it is monotone, idempotent, and satisfies $p \leq \lambda(p)$. Closure/interior operators are standard homotopy tools in finite-poset topology; the ProLT content here is the interpretation of their fixed-point regions as truth-value repair regions for a new observation.

3.3 Extremal anchors

Corollary 3.8 (Componentwise cone-anchor criterion). *[A]/[C] Suppose every connected component C of P satisfies at least one of the following:*

- (i) *C has a least element $\perp_C$ with $b(\perp_C) = 0$;*
- (ii) *C has a greatest element $\top_C$ with $b(\top_C) = 1$.*

Then refinement by ψ preserves homotopy type.

Proof. In case (i), the constant map $p \mapsto \perp_C$ on that component is a down-repair. In case (ii), the constant map $p \mapsto \top_C$ is an up-repair. The componentwise maps combine because distinct components have no comparisons. $\square$

Corollary 3.9 (Endpoint test on a Boolean-lattice signature space). *If*

$$P = 2^I$$

with its usual inclusion order, then adding one observation ψ is guaranteed to preserve homotopy whenever

$$b(\emptyset) = 0 \quad \text{or} \quad b(I) = 1.$$

Consequently, only the endpoint-reversing case

$$b(\emptyset) = 1, \quad b(I) = 0$$

can possibly change homotopy type.

Remark 3.10. This criterion is intentionally only about the canonical order-complex homotopy type. A nonredundant observation satisfying the endpoint test can still delete many order relations, alter specialization structure, and change signature distances.

4 Semilattice and Horn-type observation classes

The floor/ceiling criterion has a useful algebraic form when the realized-signature poset has meet or join structure.

Theorem 4.1 (Join-closed falsity gives a down-repair). *[S]/[A]/[C] Assume P is a finite join-semilattice. Suppose:*

- (i) *F_ψ is closed under finite joins in P ; and*
- (ii) *$F_\psi \cap \downarrow p \neq \emptyset$ for every $p \in P$.*

Then ψ has the false-floor property, with

$$\lfloor p \rfloor_F = \bigvee (F_\psi \cap \downarrow p),$$

and therefore preserves homotopy type.

Proof. The displayed join lies in F_ψ by join closure. Since p is an upper bound for every element being joined, the join is still at most p . It is clearly above every false element below p , hence is the required greatest false predecessor. $\square$

Theorem 4.2 (Meet-closed truth gives an up-repair). *[S]/[A]/[C] Assume P is a finite meet-semilattice. Suppose:*

- (i) T_ψ is closed under finite meets in P ; and
- (ii) $T_\psi \cap \uparrow p \neq \emptyset$ for every $p \in P$.

Then ψ has the true-ceiling property, with

$$\lceil p \rceil_T = \bigwedge (T_\psi \cap \uparrow p),$$

and therefore preserves homotopy type.

Corollary 4.3 (Horn-type repair). *Suppose the realized-signature poset is a meet-subsemilattice of a Boolean cube, and the truth states of ψ are the models of a Horn CNF in the old observation coordinates. If every realized signature lies below at least one such model, then ψ preserves homotopy type.*

Proof. Model sets of Horn CNF theories are closed under coordinatewise intersection, i.e. meet. The result is Theorem 4.2. $\square$

Corollary 4.4 (Dual-Horn falsity repair). *Dually, suppose P is a join-subsemilattice and the ψ -false signatures form a union-closed (equivalently, in the usual Boolean-relation setting, dual-Horn definable) family which is lower cofinal. Then ψ preserves homotopy type.*

Remark 4.5 (Why the Horn statement is not a novelty claim). Horn formulas, closure systems, and their induced closure operators are classical and highly developed. The point here is not that Horn models form closure systems; it is that those closure operators become explicit homotopy-inverse certificates for ProLT observation refinement under the stated cofinality hypotheses.

5 Height two: from a boundary matrix to a repair forest

We now obtain an exact answer in the first dimension where nonredundant homotopy-neutral refinement is possible.

Assume

$$\text{ht}(P) \leq 2.$$

Let

$$E_{\text{del}}$$

be the set of deleted edges, equivalently the b -descents, and let

$$T_{\text{del}}$$

be the set of deleted triangles. The v0.3 height-two theorem gives the relative chain complex

$$0 \longrightarrow \mathbb{F}_2[T_{\text{del}}] \xrightarrow{\partial_2^{\text{rel}}} \mathbb{F}_2[E_{\text{del}}] \longrightarrow 0.$$

5.1 The four deleted Boolean triangle patterns

For a coarse chain

$$x < y < z,$$

the surviving bit words are

$$000, \quad 001, \quad 011, \quad 111.$$

The four deleted words are:

word	number of deleted edges	deleted edges
010	1	$[y, z]$
101	1	$[x, y]$
100	2	$[x, y], [x, z]$
110	2	$[x, z], [y, z]$

Thus every column of ∂_2^{rel} has Hamming weight either one or two. That special binary fact is what turns the relative boundary matrix into an ordinary graph incidence matrix.

5.2 Definition of the repair graph

Definition 5.1 (Rooted repair graph). For each connected component C of the coarse poset P , introduce one formal root vertex $*_C$. Introduce one additional graph vertex v_e for every deleted edge $e \in E_{\text{del}}$. For every deleted triangle $\tau \in T_{\text{del}}$:

- (i) if τ contains exactly one deleted edge e , add a graph edge joining $*_C$ to v_e , where C is the coarse component containing τ ;
- (ii) if τ contains exactly two deleted edges e, f , add a graph edge joining v_e to v_f .

The resulting (possibly multi-)graph is denoted

$$\mathfrak{R}_\psi(P).$$

The one-descent triangles 010 and 101 are therefore *anchors*: they attach a descent edge directly to the component root. The two-descent triangles 100 and 110 express a dependency between two descent edges.

Proposition 5.2 (Relative boundary as reduced incidence). *[S]/[A]/[C] Over $\mathbb{F}_2$, the relative boundary matrix*

$$\partial_2^{\text{rel}}$$

is exactly the vertex-edge incidence matrix of $\mathfrak{R}_\psi(P)$ after deleting the rows corresponding to the formal roots.

Proof. A 010 or 101 triangle has one deleted edge, so its relative boundary column has one 1; the corresponding repair-graph edge joins that deleted-edge vertex to a root, whose incidence row is subsequently removed. A 100 or 110 triangle has two deleted edges, so its relative boundary column has two 1s; the corresponding graph edge joins the two deleted-edge vertices. These are all possible deleted bit patterns. $\square$

5.3 Exact repair-forest theorem

Theorem 5.3 (Height-two repair-forest theorem). *[S]/[A]/[C] Assume P is post- T_0 and $\text{ht}(P) \leq 2$. For one added observation ψ , the following are equivalent:*

- (i) *the inclusion $L_b \hookrightarrow K$ is a homotopy equivalence;*
- (ii) *it is a simple homotopy equivalence;*
- (iii) *K simplicially collapses onto L_b using only deleted edge-triangle pairs;*
- (iv) *it is a homology equivalence over $\mathbb{F}_2$;*
- (v) *∂_2^{rel} is invertible;*
- (vi) *the rooted repair graph $\mathfrak{R}_\psi(P)$ is a forest in which every connected component contains exactly one formal root.*

Thus, in height two for one new observation,

$$\boxed{\text{homology-neutral} \iff \text{homotopy-neutral} \iff \text{repair forest.}}$$

Proof. The implications (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i) $\Rightarrow$ (iv) are standard. The v0.3 height-two relative-chain theorem gives (iv) $\Leftrightarrow$ (v).

By Proposition 5.2, the matrix in (v) is a reduced incidence matrix of $\mathfrak{R}_\psi(P)$. Suppose it is invertible. Then the number of graph edges equals the number of nonroot vertices. A cycle would give a nontrivial mod-2 linear dependence among its incidence columns. A connected component containing no root would give a row dependence by summing all nonroot incidence rows in that component. A component containing two roots would give a column dependence along the unique path between those roots after the root rows are deleted. Hence the graph is acyclic and every component contains exactly one root, proving (v) $\Rightarrow$ (vi).

Conversely, suppose (vi) holds. Each rooted tree with at least one nonroot vertex has a nonroot leaf. That leaf corresponds to a deleted edge e contained in exactly one remaining deleted triangle τ . A retained triangle cannot contain a deleted edge because L_b is a subcomplex; there are no cells above dimension two. Hence e is a free face of τ , so (e, τ) is an elementary collapse pair. Removing it prunes one nonroot leaf and its incident repair-graph edge, leaving another rooted forest. Iterating removes all deleted cells and leaves exactly L_b . Thus (vi) $\Rightarrow$ (iii), closing the cycle of equivalences. $\square$

Corollary 5.4 (Greedy logical collapse algorithm). *At height two, the following algorithm decides whether one added observation preserves homotopy:*

1. *list every descent edge $p < q$ with $b(p) = 1, b(q) = 0$;*
2. *list every deleted three-chain $x < y < z$;*

3. repeatedly choose a currently deleted edge contained in exactly one currently deleted triangle and remove that edge–triangle pair;
4. succeed if all deleted cells are exhausted; fail otherwise.

The algorithm succeeds if and only if the refinement preserves homotopy.

Proof. The algorithm is exactly nonroot-leaf pruning in the repair graph. It exhausts the graph precisely when every component is a rooted tree. $\square$

Corollary 5.5 (Local repair interpretation). *A one-bit height-two refinement preserves homotopy exactly when every cluster of descent edges has:*

- (i) a path of two-descent repair dependencies leading to a one-descent anchor triangle; and
- (ii) no cyclic repair dependency.

Equivalently, every descent cluster has exactly one root anchor in the repair forest.

Remark 5.6 (Why this is stronger than the v0.3 matrix test). The v0.3 criterion required constructing ∂_2^{rel} and checking invertibility. The repair-forest theorem replaces Gaussian elimination by a purely combinatorial condition and simultaneously upgrades an exact homology test to an exact homotopy and collapse test.

6 Worked Boolean example: $\Phi = (X, Y)$

Let

$$D = \emptyset, \quad A = \{X\}, \quad C = \{Y\}, \quad B = \{X, Y\}.$$

Then

$$D < A < B, \quad D < C < B,$$

so P_Φ is the Boolean diamond B_2 . Its order complex is two filled triangles DAB and DCB glued along DB .

6.1 Positive observation: $X \vee Y$

The bit profile is

$$0, 1, 1, 1$$

on (D, A, C, B) . It is isotone, so the observation is topologically redundant. No edge or simplex is deleted.

6.2 XOR: nonredundant but homotopy-preserving

For

$$\psi = X \uparrow Y,$$

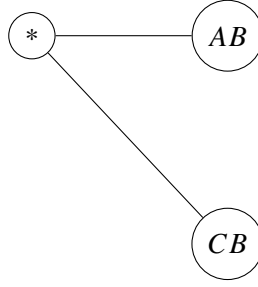
the profile is

$$0, 1, 1, 0.$$

The deleted edges are

$$A < B, \quad C < B.$$

The deleted triangles have words 010 and 010, so each has exactly one deleted edge. The repair graph is



a rooted tree. Hence

$$\mathcal{O}_{(X,Y,X \uparrow Y)} \simeq \mathcal{O}_{(X,Y)}.$$

The refinement is nevertheless nonredundant: it deletes specialization relations and changes the topology.

6.3 XNOR: another nonredundant neutral observation

For

$$\psi = X \Leftrightarrow Y,$$

the profile is

$$1, 0, 0, 1.$$

The deleted edges are $D < A$ and $D < C$. Each deleted triangle has word 101, again producing a rooted two-leaf tree. Homotopy type is preserved. The global maximum B is also a true cone anchor, so Corollary 3.8 supplies an independent certificate.

6.4 NOR: guaranteed homotopy change

For

$$\psi = \neg(X \vee Y),$$

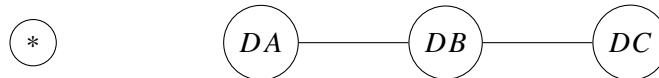
the profile is

$$1, 0, 0, 0.$$

This is nonconstant antitone. The deleted edges are

$$D < A, \quad D < C, \quad D < B.$$

The two deleted triangles both have word 100. The repair graph has an isolated root and an unrooted tree on the three descent-edge vertices:



so the rooted-forest criterion fails. Indeed the refined poset separates D from the remaining component, changing H_0 .

7 A taxonomy of logically recognizable guarantees

The results now give several increasingly broad families of homotopy-preserving observations.

Observation class / condition	Guarantee	Mechanism
b isotone on P_Φ	Topology unchanged	No descent edge exists; the observation is already open/redundant.
Down-repair exists	Homotopy preserved	Monotone weakening into a constant-false repair region.
Up-repair exists	Homotopy preserved	Monotone strengthening into a constant-true repair region.
False-floor or true-ceiling property	Homotopy preserved	Canonical interior/closure operator.
Least state false or greatest state true in each component	Homotopy preserved	Each refined component remains a cone.
Meet-closed, upper-cofinal truth set	Homotopy preserved	Closure operator; includes Horn-type cases on meet-semilattice signatures.
Join-closed, lower-cofinal false set	Homotopy preserved	Interior operator; dual closure-system criterion.
One bit over a coarse chain, except nonconstant nonincreasing threshold	Homotopy preserved	Exact v0.3 beat-point classification.
Height two and repair graph is rooted forest	Homotopy preserved, exactly	New exact repair-forest theorem; explicit simplicial collapse.
Nonconstant antitone bit on connected coarse component	Homotopy not preserved	All cross-level comparisons disappear; H_0 changes.
Height at most one, nonredundant refinement	Homotopy not preserved	Any deleted graph edge changes Euler characteristic/homology.

8 Beyond height two: discrete Morse repair

The repair-forest theorem works because, in dimension two and with one new bit, every deleted triangle has one or two deleted edges. In higher dimension the natural replacement is relative discrete Morse theory.

Definition 8.1 (Complete descent matching). Let

$$\mathcal{D}_b := K \setminus L_b$$

denote the set of deleted simplices. A *complete descent matching* is a matching on the relative face poset of deleted simplices in which every deleted simplex is paired with a deleted face or coface of adjacent dimension, and the matching is acyclic in the sense of discrete Morse theory.

Theorem 8.2 (General discrete-Morse preservation criterion). *[S]/[A] If the deleted simplices admit a complete acyclic relative matching, then*

$$K \searrow L_b$$

through a sequence of simple-homotopy reductions, and therefore adding ψ preserves homotopy type.

Remark 8.3. At height two, the rooted repair forest supplies exactly such a complete matching by repeatedly pairing a nonroot leaf descent-edge with its unique incident deleted triangle. Thus Theorem 5.3 can be viewed as a fully solved low-dimensional instance of the general discrete-Morse repair problem.

A promising higher-dimensional target is therefore not to search for an impossible universal homotopy classifier, but to identify logical observation classes for which a canonical first-descent or repair-state matching is automatically acyclic.

9 Computational verification

An independent verifier accompanies this note. It constructs finite posets directly, applies one-bit post- T_0 refinement, builds order complexes, relative boundary matrices, and the rooted repair graph, and checks the theorems without assuming their proofs.

9.1 Repair-map checks

Across every naturally labelled transitive poset on at most five vertices and every one-bit labeling, the verifier checked:

- (i) every componentwise anchor-safe observation is homology-neutral;
- (ii) every false-floor observation is homology-neutral;
- (iii) every true-ceiling observation is homology-neutral.

No counterexample occurred.

9.2 Height-two exhaustive repair-forest check

For every naturally labelled height-two poset through six vertices and every one-bit observation, the verifier checked the exact equivalence

$$H_*(K, L_b; \mathbb{F}_2) = 0 \iff \mathfrak{R}_\psi(P) \text{ is a rooted forest.}$$

The counts were:

n	height-two posets	one-bit cases	neutral nonredundant	repair-forest cases
3	1	8	2	2
4	13	208	50	50
5	172	5,504	1,232	1,232
6	2,787	178,368	36,783	36,783

Across $n \leq 6$, this is 184,088 height-two one-bit cases, including 38,067 nonredundant neutral refinements. Every one of those 38,067 cases had the rooted-forest collapse certificate. The computational result is a verification of the theorem on small finite instances, not the proof; the proof is Theorem 5.3.

10 Prior art and novelty boundary

Several mathematical ingredients used here are standard:

- (i) finite-poset order complexes and Quillen/McCord fiber methods;
- (ii) pointwise homotopies of monotone poset maps and closure/interior operators;
- (iii) simplicial collapse and discrete Morse matchings;
- (iv) graph incidence matrices and the characterization of trees by invertible reduced incidence matrices;
- (v) Horn logic as a closure-system formalism;
- (vi) topology associated with Boolean functions more broadly.

In particular, Barmak proves strong simple-homotopy versions of Quillen-type poset fiber theorems [2]. Forman and later discrete-Morse treatments provide the collapse/matching machinery [4, 5]. Wild surveys the closure-system viewpoint on pure Horn formulas [6]. Björner, Goresky, and MacPherson explicitly study order complexes associated with Boolean functions [7]. Therefore no novelty should be claimed for “Boolean logic plus topology,” closure operators, or discrete Morse theory as such.

The contribution candidates requiring a focused novelty audit are instead:

- (a) the ProLT-specific repair-map formulation of homotopy-preserving observation refinement;
- (b) the exact rooted repair-forest representation of the height-two one-bit relative boundary;
- (c) the equivalence of height-two one-bit homology neutrality, homotopy neutrality, simple collapse, and the rooted-forest condition;
- (d) the greedy descent-edge repair algorithm interpreted directly in terms of logical observation values;
- (e) integration of these criteria with the earlier refinement-defect and evolving-ProLT framework.

11 Next research targets

The strongest next directions are now fairly precise.

- N1. Multi-bit height-two refinement.** A block of new observations can make a deleted triangle contain one, two, or three deleted edges. The repair graph therefore becomes a rooted hypergraph. Determine the correct acyclicity/matroid condition replacing the tree theorem.
- N2. Higher-dimensional first-descent matching.** For one bit in arbitrary height, construct canonical relative Morse matchings from the first $1 \rightarrow 0$ descent of a deleted chain. Classify posets/formulas for which the matching is complete and acyclic.
- N3. Logical fragments.** Systematically test Horn, dual-Horn, bijunctive, affine/XOR, monotone, and unate observation formulas on structured signature posets. Determine which fragments admit polynomial-time homotopy-preservation certificates.

- N4. Persistent refinement defects.** Along a post- T_0 refinement sequence, record when repair forests cease to exist and when relative critical cells first appear. This can supply event times for the refinement axis of the two-parameter persistence program developed in v0.3.
- N5. Adversarial novelty audit.** Search finite-poset topology, Boolean-function topology, closure systems, SAT topology, discrete Morse theory, and knowledge-representation literature specifically for constructions equivalent to the repair graph/forest theorem before making any publication claim.

12 Conclusion

There is now a useful hierarchy of answers to the question “which new logical observations preserve homotopy?”

At the easiest level, isotone observations are redundant. More generally, monotone down- or up-repair maps provide homotopy inverses, with false floors, true ceilings, cone anchors, and semilattice closure systems as concrete subclasses. Horn-type closure systems fit naturally into this repair picture under cofinality hypotheses.

The height-two one-bit case is stronger: it is completely solved by a rooted graph. A logical observation preserves the homotopy type exactly when its descent edges and deleted three-chains assemble into a rooted forest. When they do, the forest directly prescribes the elementary collapse sequence. Thus the new observation may substantially refine the logical topology while leaving the canonical homotopy type unchanged, and the reason can be read combinatorially from its truth pattern rather than inferred only after a homology computation.

A Verification artifacts

The source distribution includes:

1. `verify_homotopy_preserving_observations.py`, checking repair-map, anchor, floor/ceiling, and perfect-matching criteria on small finite posets;
2. `verify_repair_forest_theorem.py`, independently constructing the rooted repair graph and testing the exact height-two theorem through six vertices;
3. `height2_repair_forest_summary.csv`;
4. `homotopy_guarantee_coverage.csv`.

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